

Offset Loading in a Bilateral Squatting Movement Pattern Influences Ground-Reaction Force and Muscle Activity in the Dominant and Nondominant Limb

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Purpose: The purpose of this study was to explore whether offset loading in the barbell squat altered ground-reaction force (GRF) and muscle activation in the dominant (D) and nondominant (ND) lower limb compared to traditional squats. **Methods:** Twelve well-trained men (age 26.4 [3.2] y; 10.3 [1.9] y experience) performed 3 sets of 10 repetitions at 60% of their previously measured 1-repetition maximum. Sets were quasi-randomized between traditional loading (TDL), dominant-side offset loading (OS-D), and nondominant-side offset loading (OS-ND). All repetitions were performed on a dual force plate with electromyography sensors on the prime mover muscles of the squat. GRF symmetry was assessed using the symmetry index (SI) to determine the direction (D [+] or ND [-]) and magnitude (%) of the asymmetry. Finally, the first 3 and final 3 repetitions of each set were compared for compensatory changes in symmetry. **Results:** OS-D induced a significant change in limb SI relative to TDL (5.21% vs 1.44%; $P = .011$); however, no significant difference in limb SI was seen between TDL and OS-ND (-0.66% vs 1.44% ; $P = .278$). No asymmetries between D and ND muscle activation were present in any condition. TDL and OS-D squats exhibited significant improvements in limb SI between the first 3 and final 3 repetitions ($P = .035$ and $.011$, respectively); however, no such improvement was seen in OS-ND. **Conclusions:** OS-D is capable of significantly altering GRF limb SI in a bilateral squat; however, OS-ND appears to exhibit no GRF or electromyography effects relative to TDL. Thus, the results of this study do not support the use of OS-ND in the pursuit of strengthening a weaker limb, suggesting that unilateral training may be a preferred mode of exercise for this desired outcome.

Keywords: symmetry, asymmetry, strength training, sport performance

Limb dominance is a trait found in numerous species across the globe.¹ Human handedness is thought to have developed in pre-*Homo* species,² but clearer evidence is often traced to more recent ancestors via specific tool designs and other archaeological artifacts.³ The preferred use of 1 limb over another originates from lateralization in the sensorimotor cortex⁴ and tends to appear between 7 and 13 months of age.⁵ It has been suggested that handedness directs which limb is used for complex tasks and which is utilized for support roles.^{1,6} In the lower body, the dominant limb (D) may be described as the limb that is preferred for manipulating objects (eg, striking a ball) or which limb leads in a jumping motion, whereas the nondominant (ND) lower limb is described as the extremity that provides support for these tasks.⁷ In addition, environmental factors, such as repetitive motor tasks, can influence limb dominance as well as the magnitude of strength asymmetries between limbs.⁸ Thus, others have designated dominance as the limb that attains better results in performance-related assessments.⁸ In the strength and conditioning field, these definitions have led to researchers exploring how limb dominance and strength asymmetries may impact sport performance and injury risk.

The concept of limb dominance and its associated motor control patterns inherently induce a greater demand and load on the D limb throughout life.⁹ Over time, some have opined that this predilection results in substantial limb strength asymmetries between D and ND limbs and that these asymmetries may

negatively influence sport performance and injury risk.¹⁰ Curiously, inconsistent findings of the association between asymmetries and sport performance have been reported.⁸ These discrepancies are likely influenced by the sporting history and training experience of the subjects being studied.⁸ Moreover, perceived limb dominance also may not always predict performance differences between limbs, thus the definition of limb dominance in studies may vary between subjective perception and objective data.⁸ Regardless, researchers and practitioners have sought methods of mitigating these asymmetries, but investigators are currently faced with the uncertainty of this desired outcome. The motor control commands that underpin limb dominance are likely permanent in nature¹¹ and, thus, training interventions to promote limb strength symmetry and reduce the relative daily load on the dominant limb may be performed in vain.

One recent strategy that has been proposed to reduce limb strength asymmetries, especially during resistance training, is offset loading. Offset loading is the practice of applying an uneven load to a resistance training implement (eg, barbell), thus exposing 1 limb to greater resistance than the other during a bilateral movement. The initial studies performed on offset loading investigated the effects of placing a predetermined portion of the training load (2.5%–10%) on either the D or ND side of the barbell during the bench press movement. Jarosz et al¹² reported that adding as little as 2.5% of the training load to the ND side resulted in more symmetrical muscle activation of the prime mover muscles when compared to traditional (evenly loaded) bench press. In isolation, these data suggest that

symmetrical muscle activation may be achieved in bilateral strength training movements by using offset loading, however, the long-term neuromuscular adaptations of this practice are unknown. Moreover, the extent of offset loading literature has only examined the bench press whereas the majority of limb dominance and sport performance data have been collected from the lower body limbs.

Therefore, exploring offset loading in lower body strength training movements, such as the barbell squat, may offer additional insight to practitioners in regard to the acute effects of offset loading. Thus, the purpose of this study was to investigate acute changes in ground-reaction force (GRF) and muscle activation during offset loading in the barbell squat to determine whether or not offset loading induces acute alterations in GRF or muscle activation. Given that previous studies have reported acute changes in prime mover muscle activity during offset loading in the barbell bench press,^{12,13} it was hypothesized that offset loading would increase limb GRF and muscle electromyography (EMG) in the side resisted by the heavier load. In addition, we hypothesized that offset loading toward the ND limb (OS-ND) would produce more symmetrical GRF and EMG than traditional (TDL) loading.

Methods

Subjects

Following an a priori power analysis (power = 0.83), 12 healthy men aged 18–35 with at least 1 year of squatting experience were recruited for this study (Table 1). Inclusion criteria included: male, 18–35 years, 1+ years of squatting experience, estimated squat 1RM > 1.5× bodyweight. Potential subjects were sent a questionnaire outlining their availability for participation, recent (1 y) injury history, and an estimation of their current squat 1RM. If subjects were available for participation, healthy, and met the minimum strength and experience requirements, they were invited for participation. Following institutional review board approval, all subjects signed an informed consent form before participating.

Study Design

This study was a crossover experimental design. One subject group was recruited for participation and all subjects underwent baseline testing followed by a single experimental session, including all 3 conditions. The first session involved baseline assessment of the subject’s symmetrically loaded squat 1RM as well as familiarization with offset loading. In addition, subjects were asked to determine their D and ND lower limb by identifying the preferred limb used for kicking tasks.¹⁴ Approximately 1 week after the baseline test, subjects returned for the experimental session. This

session included assessment of squat maximal voluntary isometric contraction (MVIC) EMG as well as muscle activation on the squat with randomized offset loads. Following the MVIC assessment, subjects underwent the experimental protocol.

For the experimental protocol, subjects followed a standardized warm-up protocol and then performed 3 sets of 10 repetitions at 60% of their previously determined squat 1RM on a dual GRF plate with EMG sensors on the prime mover muscles in the squat. The order of these sets was quasi-randomized between TDL, dominant side offset (OS-D), and nondominant-side offset (OS-ND). Three minutes of rest were taken between each set to ensure full recovery between sets. Offset conditions employed a 5% offset with the greater load placed on the assigned side (OS-ND or OS-D). For example, a prescribed load of 60% of a 100 kg 1RM would be 60 kg. The adjustable load is the prescribed load (60 kg) minus the barbell mass of 20.4 kg, equaling 39.6 kg of adjustable load. Adjusting to the balanced load used in TDL would amass 19.8 kg on each side. In offset loading, a 39.6 kg adjustable load would create a loaded side of 20.8 kg and a deloaded side of 18.8 kg, accounting for 52.5% and 47.5% of the adjustable load, respectively. All loads were rounded to the nearest 0.2 kg.

Squat 1RM

Symmetrically loaded squat 1RM was determined during the first laboratory visit. All participants performed a standardized warm-up that included 3 minutes on a cycle ergometer (Rogue Fitness) at 140 Watts followed by a general dynamic warm-up that involved movements for both the lower and upper body. Following the general warm-up, participants underwent a specific warm-up on the squat according to previously published National Strength and Conditioning Association (NSCA) guidelines.¹⁵

Squat stance width foot positions were marked and maintained for both the MVIC and experimental session. Participants were instructed to unrack the barbell along their trapezius muscle and take 2 to 3 steps back with each foot. Once the participant was settled, they descended into the squat. A successful attempt required the top of the thigh to break the parallel plane with the floor and the subject to return to the start position with no assistance from the spotter. Subjects were allowed to perform the movement at their self-selected tempo.¹⁵ Two experienced spotters (NSCA Certified Strength and Conditioning Specialists) were present for all attempts to ensure safety and judge successful attempts. All 1RM testing was performed using a 20.4 kg power barbell (Texas Power Bar) and standard power rack (Prime Fitness). No assistance equipment, such as a belt, knee wraps, knee sleeves, and so on, was used by the participants. All participants achieved their 1RM within 4 attempts.

Following squat 1RM assessment, participants were given a 5-minute rest and then underwent the familiarization protocol. Subjects performed 2 sets of 5 repetitions at 60% of their 1RM. Both offset sets involved a 5% offset load and 2 minutes of rest were given between each set.

Electromyography and Squat MVIC

An 8-channel Delsys Trigno Avanti system (Delsys Inc) was used for recording and analyzing electrical activity from the primary D and ND muscles involved in the squat, the vastus lateralis, vastus medialis, and erector spinae. Prior to placing electrodes on each muscle group, the placement site was identified according to SENIAM (Surface Electromyography for the Non-Invasive Assessment of Muscles) guidelines,¹⁶ cleaned, shaved, and cleaned again using alcohol wipes. Electrodes were placed in parallel with the

Table 1 Descriptive Characteristics, N = 12

Variable	Average (SD)
Age, y	26.4 (3.2)
Height, m	1.81 (0.06)
Body weight, kg	86.8 (12.6)
1RM, kg	160.6 (23.1)
1RM:BM, kg/kg body weight	1.87 (0.29)
Experience, y	10.33 (1.92)

Abbreviations: BM, body mass; 1RM, 1-repetition maximum.

presumed underlying muscle fiber arrangement in each muscle group. Signals were then band-pass filtered with cutoff frequencies of 20 to 450 Hz. Following data collection, EMG signals were analyzed via EMGworks software from which EMG root mean square was calculated.

MVIC was measured at least 72 hours after squat 1RM. Participants performed the general warm-up from the 1RM testing session and were then given 2 MVIC attempts of 50% effort and 75% perceived effort. Following the MVIC warm-up, subjects performed 2 maximal MVIC attempts. Each MVIC attempt lasted 5 seconds, and 30 seconds of rest were given between the final maximal 2 attempts. The highest value in the final 2 attempts was recorded as 100% MVIC through which the experimental EMG outcomes could be expressed.

The position for the MVIC involved an isometric squat performed with 90 degrees of knee and hip flexion. This position was selected as previous research has demonstrated that a knee angle of 90° produced the greatest muscle activation in the isometric squat when compared with knee angles of 20° and 140°. ¹⁷ Participants set up in a standard power rack (Prime Fitness) with a 20.4 kg power bar (Texas Power Bar) resting on safety spotter arms just below the MVIC position. Two additional spotter arms were placed 15 to 18 cm above the bottom spotter arms dependent on the subject's limb lengths and what position allowed for the previously mentioned knee and hip flexion angles. Following a verbal countdown and start command from a researcher, participants explosively squatted the barbell into the second pair of safety spotter arms and continued pushing with a maximal effort for 5 seconds. EMG signals for the D and ND vastus lateralis, vastus medialis, and erector spinae were recorded during this 5-second period, and peak signals (from either attempt) were recorded for each muscle group. The highest value in the final 2 attempts was used for normalizing % MVIC during the experimental protocol.

Ground-Reaction Force

Vertical GRF of the D and ND limbs during dynamic squats was analyzed using a dual force plate (Leonardo Mechanograph GRFP XL, Novotec Medical GmbH). The platform contained 2 symmetrical force plates that separate the device into left and right halves. Each half contained 4 strain gauge force sensors which are connected to a laptop computer via a USB 2.0 connection. The signals from the force sensors were sampled at a frequency of 800 Hz and analyzed using the Mechanography GRFP Research Edition software. Peak GRF for the D and ND limbs was averaged for all 10 repetitions in each condition and analyzed for symmetry within and between conditions. Limb symmetry was assessed using the limb symmetry index (SI). The limb SI was calculated as

$$\frac{\text{GRF (D)} - \text{GRF (ND)}}{\text{GRF (D)}} \times 100.$$

Table 2 Limb SI Within and Among Conditions

Condition	Peak ground-reaction force				D vs ND ES (Cohen d)
	D (N)	ND (N)	P, D vs ND	SI (%) (95% CI)	
Traditional	1179 (207)	1167 (162)	.524	1.44 (4.9) (−1.33 to 4.21)	0.06
OS-D	1197 (194)	1149 (172)	.084	5.21 (4.86)*, + (2.46 to 7.49)	0.26
OS-ND	1178 (195)	1185 (167)	.932	−0.66 (4.25) (−3.06 to 1.74)	0.03

Abbreviations: D, dominant limb; ES, effect size; ND, nondominant limb; OS-D, offset-dominant; OS-ND, offset-nondominant; SI, symmetry index.

*Significantly different ($P < .05$) than TDL. +Significantly different ($P < .05$) than OS-ND.

Experimental Session

Following MVIC testing, participants performed the experimental session to ensure electrode placement remained consistent between the MVIC and dynamic repetitions. The standardized specific warm-up was performed 5 minutes after the MVIC session. Following the warm-up, participants underwent 3 randomized sets of 10 repetitions at 60% of their symmetrically loaded squat 1RM with 3 minutes of rest between sets. One set was traditionally loaded (TDL), whereas the other 2 sets included a 5% offset load on either the D (OS-D) or ND (OS-ND) side to the closest 0.2 kg.

Statistical Analysis

All data are reported as mean (SD). A 2×3 (side $\times$ condition) repeated-measures analysis of variance (ANOVA) was performed to analyze differences in average peak GRF between D and ND limbs within conditions, whereas a 1×3 (SI $\times$ condition) repeated-measures ANOVA was employed to detect differences in average limb SI among conditions. Differences in average muscle activation between D and ND muscles within conditions were analyzed via a 2×3 repeated-measures ANOVA (muscle $\times$ condition) for each muscle pair. In addition, a repetition phase interaction for GRF was explored. Two phases (first 3 repetitions and final 3 repetitions) of each condition were examined using a 2×3 (phase $\times$ condition) repeated-measures ANOVA. Bonferroni post hoc analysis was employed in the case of any significant effects. An a priori alpha level of .05 was utilized, and all statistical analyses were performed using SPSS (version 28.0, SPSS Inc).

Results

Force and Symmetry

Limb force and symmetry findings are presented in Table 2 and repetition comparisons are shown in Figure 1A–1C. It was found that OS-D had a significantly higher limb SI than both TDL ($P = .006$) and OS-ND ($P < .001$), whereas no difference was detected between TDL and OS-D ($P = .278$). Figure 1A–1C show the results of paired t tests between D and ND limb for each repetition within each condition.

Muscle Activation

Within conditions muscle activation data are presented in Table 3. No significant differences ($P < .05$) were discovered within or among any conditions.

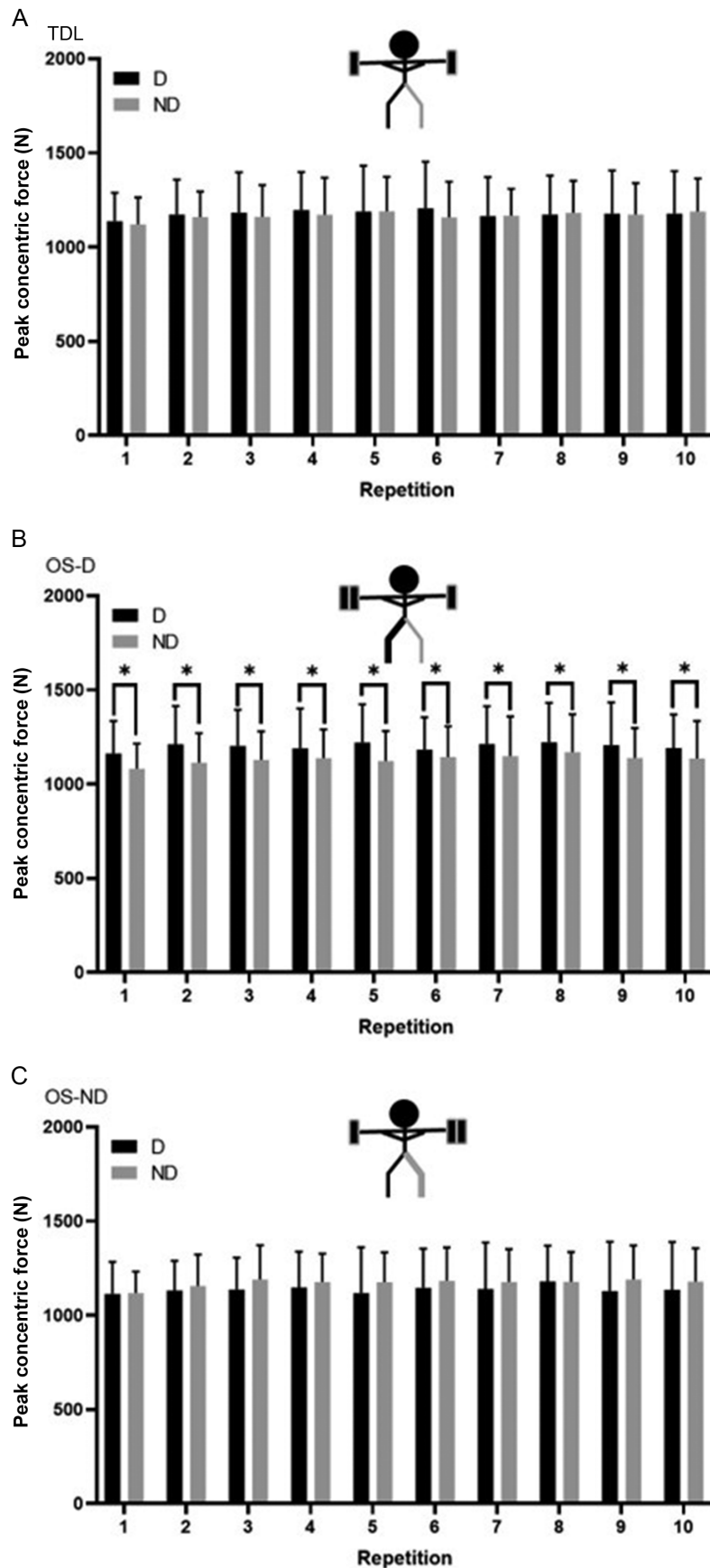


Figure 1 — Limb ground-reaction-force symmetry per repetition in (A) TDL, (B) OS-D, and (C) OS-ND. D indicates dominant limb; ND, nondominant limb; OS-D, dominant-side offset loading; OS-ND, nondominant-side offset loading; TDL, traditional loading. *Significant difference ($P < .05$) between limbs.

Table 3 D and ND Muscle Activation Within Conditions

Condition	Muscle	D (%MVIC) (95% CI)	ND (%MVIC) (95% CI)	P	D vs ND effect size (Cohen d)
Traditional	VM	58.5 (19.9) (47.2 to 69.8)	49.9 (23.5) (36.6 to 63.2)	.229	0.39
	VL	61.1 (18.5) (50.6 to 71.6)	50.7 (27.7) (35.0 to 66.4)	.160	0.44
	ES	50.1 (22.5) (37.3 to 62.8)	49.8 (21.4) (37.7 to 61.9)	.985	0.01
OS-D	VM	55.0 (17.8) (44.9 to 65.1)	47.1 (16.0) (38.0 to 56.2)	.257	0.47
	VL	58.5 (18.6) (48.0 to 69.0)	50.4 (26.2) (35.6 to 65.2)	.313	0.36
	ES	53.3 (28.7) (37.1 to 69.5)	46.9 (26.7) (31.8 to 62.0)	.685	0.23
OS-ND	VM	58.9 (19.2) (48.0 to 69.8)	51.3 (25.7) (36.8 to 65.8)	.322	0.33
	VL	58.9 (16.5) (49.6 to 68.2)	55.9 (27.2) (40.5 to 71.3)	.684	0.13
	ES	56.2 (22.7) (43.4 to 69.0)	42.9 (22.4) (30.2 to 55.6)	.383	0.59

Abbreviations: D, dominant; ES, erector spinae; %MVIC, % of maximal voluntary isometric contraction; ND, nondominant; OS-D, offset-dominant; OS-ND, offset-nondominant; VL, vastus lateralis; VM, vastus medialis.

Phase Analysis

For GRF, a significant reduction was found between phases in TDL and OS-D but not in OS-ND (Table 4).

Discussion

The aim of this study was to investigate the acute effects of offset loading in the barbell squat. It was found that using an offset load on the D side can significantly increase GRF in the D limb relative to the ND limb. However, no such effect was seen with offset loading on the ND side. In addition, a compensation effect was found during TDL and OS-D in which limb SI significantly improved throughout the set. Again, no such effect was seen in OS-ND, which suggests that, in this cohort of subjects, the acute response to OS-ND exhibits no differences from TDL. These findings stand in modest contrast to previous offset loading literature which reported acute changes in prime mover activation during offset loading in the barbell bench press.^{12,13}

Previous research has reported significant force asymmetries between D and ND limbs during barbell squats, with limb symmetry indexes ranging from 2.09% to 5.94%.^{18–20} In this study, an average limb SI of 1.44% (4.9%) was found during TDL squats. It is conceivable that the training status of the subjects in this study (>10 y of experience) influenced the visually lower limb SI in TDL compared with prior studies.²¹ However, during offset loading, OS-D achieved a significantly higher limb SI than TDL (5.21% [4.86%]; $P = .011$), whereas OS-ND was not statistically different from TDL (−0.66% [4.25%]; $P = .278$). The alteration in the balance of limb GRF is likely predicated by the placement of the offset load. Since GRF is a measure of an equal and opposite reaction to the vertical force produced by each limb,²² it is possible to deduce that an increased load on one side of the barbell necessitated greater force development by the same side limb. Despite this potential causality, the alteration in limb GRF only achieved statistical significance in OS-D but not OS-ND. It is conceivable that the sensory feedback involved in motor control is less substantial during OS-D than OS-ND. Since reliance on the D limb is a natural human trait, shifting additional load to the ND limb during OS-ND may provoke additional afferent feedback as the increased ND load may be sensed as a deviation that interferes with the motor task.^{23,24} Thus, a potential compensation may be immediately adopted in OS-ND, leading to symmetrical GRF between limbs whereas no such compensation occurs during OS-D. Similarly, subjects with previous ACL (anterior cruciate ligament)

Table 4 Ground-Reaction-Force Limb SI Due to Phase Within Condition

Condition	Phase	Limb SI (%)	Δ , %	P
TDL	R _{1–3}	2.39 (5.37)	−143.93*	.035
	R _{8–10}	−1.05 (5.83)		
OS-D	R _{1–3}	6.69 (5.91)	−47.68*	.011
	R _{8–10}	3.50 (4.25)		
OS-ND	R _{1–3}	−1.24 (5.22)	+149.19	.318
	R _{8–10}	0.61 (6.58)		

Abbreviations: Δ , relative change between phases; Limb SI, average symmetry index within phase; OS-D, offset-dominant; OS-ND, offset-nondominant; TDL, traditional.

*Significant difference ($P < .05$) between phases.

reconstruction surgery have been shown to adopt compensation patterns during the squat motion that favor unloading of the previously injured limb,²⁵ thus, bilateral squatting patterns may inherently rely on force production from the stronger or dominant limb. While the data are limited, especially in healthy subjects, it is possible that OS-ND invokes a greater motor feedback response which directs compensation patterns to unload the ND limb despite the increased offset load.

Shifts in limb SI and alterations in GRF were not accompanied by significant changes in muscle activation. The inspiration for this study was derived from previous offset loading works^{12,13} that reported alterations in muscle activation with offset loading in the bench press. Specifically, the muscles resisted by the heavier load exhibited greater rates of activation than their less-resisted counterparts. However, in this study, no EMG asymmetries between D and ND muscles were present within or among any condition. The relative offset load utilized in this study (5%) was similar to the offset loads used in previous investigations (2.5%–10%),^{12,13} thus, the primary difference between this study and previous publications was the use of the squat, rather than the bench press. One potential mechanism that may inhibit significant changes in lower body EMG, especially during offset loaded squats, is the vestibular system. The vestibular system, in combination with somatosensory receptors, seeks to maintain posture and balance during upright tasks.²⁴ Indeed, a barbell squat may invoke perturbations through trunk sway that lead to vestibular sensory feedback, which can alter motor commands according to the task at hand.^{24,26} Since achieving a stable and symmetrical posture is a primary function of the

vestibular system,²⁷ symmetrical motor commands may be one method of achieving this outcome. It is plausible, then, that any form of squatting may inherently require more symmetrical muscle activation between D and ND limbs due to the required maintenance of an upright and balanced posture. Moreover, well-trained athletes exhibit greater balance during perturbations than nonathletic subjects,²⁸ which suggests that training status may also affect the ability of offset loading to influence acute changes in GRF or EMG. With these concepts in consideration, offset loading in the barbell bench press may be able to invoke acute changes in muscle activity whereas such alterations are not seen in the barbell squat. However, these theories certainly require further exploration.

This vestibular feedback system may also help explain the shift in limb SI during the phase analysis. In the TDL and OS-D condition, subjects exhibited significant improvements in limb SI between the first 3 and last 3 repetitions of the condition. It is well understood that adaptations to resistance training can occur both acutely and chronically. Acute adaptations are thought to be executed within the central nervous system and may be visualized by improvements in movement efficiency and execution²⁹ and these immediate adaptations may be influenced by vestibular feedback.^{24,26} It is likely that the subjects in this study may have exhibited improvements in movement efficiency throughout a resistance training set, which was reflected by the reduction of limb GRF asymmetry during TDL and OS-D squats. Previous research has reported similar findings in recreationally trained subjects. Hodges et al¹⁸ analyzed absolute limb GRF asymmetry in the first 2 and final 2 repetitions within a set of 8 repetitions at 90% of their 8RM. Curiously, when the researchers removed highly asymmetrical subjects from their analysis, the average limb SI was significantly reduced between the first 2 and last 2 repetitions ($P = .044$). These preliminary data suggest that acute neuromuscular and biomechanical adaptations to a squatting exercise may occur within a single set, and that these adaptations appear to favor an approach toward symmetry. Despite this theory, there was no improvement in limb SI seen in the OS-ND condition.

Practical Applications

While offset loading appears to potentially have use in the bench press, as demonstrated by Jarosz et al,¹² the results of this study indicate that the acute effects of offset loading in the squat are less clear. Indeed, no acute changes in EMG or GRF were uncovered during OS-ND which renders the practice of OS-ND potentially futile in regard to diminishing asymmetries between D and ND limbs. Moreover, OS-D may serve to further strengthen the D limb compared with the ND limb, thus expanding a strength deficit between limbs. Currently, practitioners seeking to reduce limb strength asymmetries may be better served by employing unilateral exercises as a primary stimulus for adaptation as these approaches appear to be more effective at preferentially strengthening a weaker limb.³⁰ Further research should endeavor to understand the acute effects of offset loading in individuals with known asymmetries (injury rehabilitation, etc) as well as the long-term adaptations to offset loading.

Limitations

As with any strength training study (acute or longitudinal), the cohort size ($n = 12$) is a limitation. While an a priori power analysis suggested this to be a sufficient population (power = 0.83), a larger number of participants would almost certainly provide clearer

outcomes. Moreover, highly trained subjects were utilized in this study, thus confounding potential limb strength asymmetries. It is uncertain as to whether or not these results would be replicated in other populations, especially those with less training experience or those with preexisting known asymmetries. The highly experienced nature of the participants also allowed for greater ecological validity when applying to athletic populations, thus the authors chose to allow participants to select their own lifting pace to better reflect field application. However, it is plausible that the use of maximal intended velocity may have generated more robust EMG signals which could have further clarified the muscle activation data. Despite this theory, previous research has reported that subjects using a self-selected repetition tempo displayed greater muscle activation than performing designated repetition tempos³¹ which may suggest that our lack of maximal intended velocity was not a limitation. Finally, employment of greater offset loads may invoke more robust changes in GRF and EMG, however, only 5% was used in this study. Pilot testing with 7.5% and 10% offset loads resulted in imbalances that were too substantial for the subjects to maintain control of the barbell and, thus, these offset loads were not explored further.

Conclusion

The results of this study suggest that offset loading may shift limb SI toward the limb resisted by the heavier load. While this intuitively increases GRF in the target limb, only OS-D displayed a significant change in limb SI. Furthermore, offset loading does not appear to influence quadriceps or erector spinae muscle activity to a significant degree. Thus, the potential long-term symmetry and performance adaptations to offset loading are difficult to predict, especially with the lack of alterations in GRF and EMG seen in OS-ND. There did appear to be an acute compensation in force output in TDL and OS-D in which symmetry improved throughout the set, whereas no such effect was seen in OS-ND. This indicates that acute neuromuscular adaptations may allow a trainee to compensate for offset loads on the D side, but not ND side. The ability of this acute phenomenon to predict long-term neuromuscular function for athletic performance or injury prevention is unknown.

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